**Experiment Number: 2**

**MATLAB Simulation of Cellular Coverage, Cell Splitting and Handoff in a Multi-Cell Network**

**Aim**

**To model a simplified hexagonal cellular network and study how cell radius affects coverage,**

**estimated user capacity and handoff as a mobile crosses a cell boundary.**

**Objectives**

**By the end of the experiment, students should be able to:**

**1. Model adjacent hexagonal cell coverage and identify cell centers and boundaries.**

**2. Estimate the number of cells required to cover a fixed service area for different cell radii.**

**3. Compare estimated users served under an explicit per-cell capacity assumption.**

**4. Simulate a mobile moving across a cell boundary and identify the serving-cell transition.**

**Requirements**

- **Computer with MATLAB R2019b or later.**
- **No RF hardware is required; this is a geometric and analytical simulation.**

Theoretical Background

A regular hexagonal cell is an idealized coverage region. For cell radius R, measured from

the center to a vertex, its area is $A_{cell} = (3\sqrt{3}/2)R^2$. A first-order estimate for a service area

A is $N_{cells} = \text{ceil}(A/A_{cell})$. This area ratio ignores boundary overlap, roads, terrain,

irregular coverage and the fact that real cell layouts extend beyond the service boundary.

For a simple capacity comparison, assume each cell can serve C users at the same time.

Estimated network capacity is $N_{cells} \times C$. This is a planning illustration, not a prediction of

a real network; capacity also depends on spectrum, scheduling, traffic, interference, backhaul

and radio conditions.

For handoff, the serving cell is approximated as the cell center nearest to the mobile. A

handoff is indicated when the nearest-center identity changes. Real systems use measured

radio quality, hysteresis, time-to-trigger, load and network policy, so this geometric rule is

only a teaching model.

**Parameters and Assumptions**

| Parameter | Baseline value | Meaning |
|---|---|---|
| Service area | 5 $km^2$ | Area to be covered by the idealized layout |
| Cell radii | 1.0 km and 0.5 km | Baseline and split-cell cases |
| Per-cell simultaneous-user capacity | 30 users/cell | Same assumed capacity per cell in each case |
| Handoff geometry | Two adjacent cells | Mobile follows a straight path across their shared boundary |

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Record any values changed. The capacity comparison holds the per-cell capacity constant to

isolate the effect of cell count; it does not imply that smaller real cells always have exactly

this capacity.

**Procedure**

1. Run the MATLAB script below and record the cell area, estimated cell count and total

illustrative capacity for each radius.

2. Inspect the hexagonal layout and confirm that adjacent cells share edges without gaps in

the idealized tiling.

3. Inspect the mobile-path plot and note the simulated handoff point.

4. Change the service area, radii or users-per-cell assumption one at a time; rerun and

compare results.

5. Explain why the geometric cell count and nearest-center handoff differ from an operational cellular network.

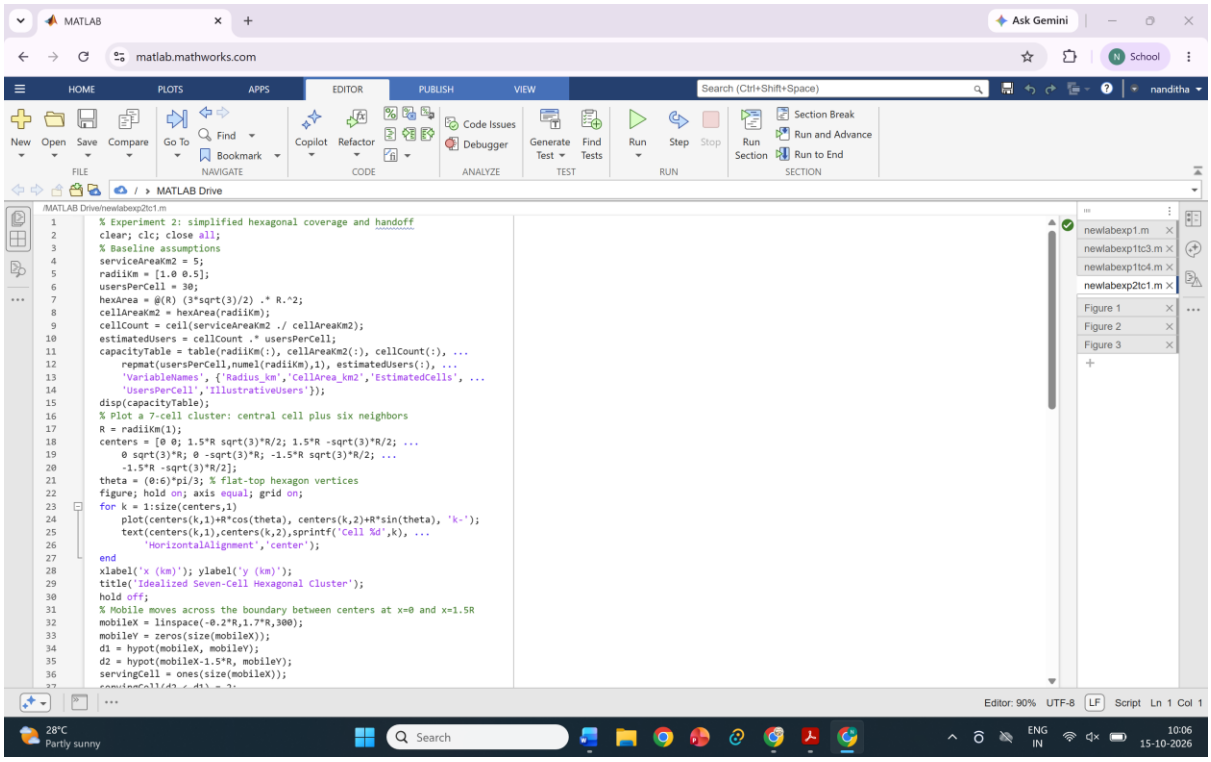
**MATLAB Program — Coverage, Cell Splitting and Handoff**

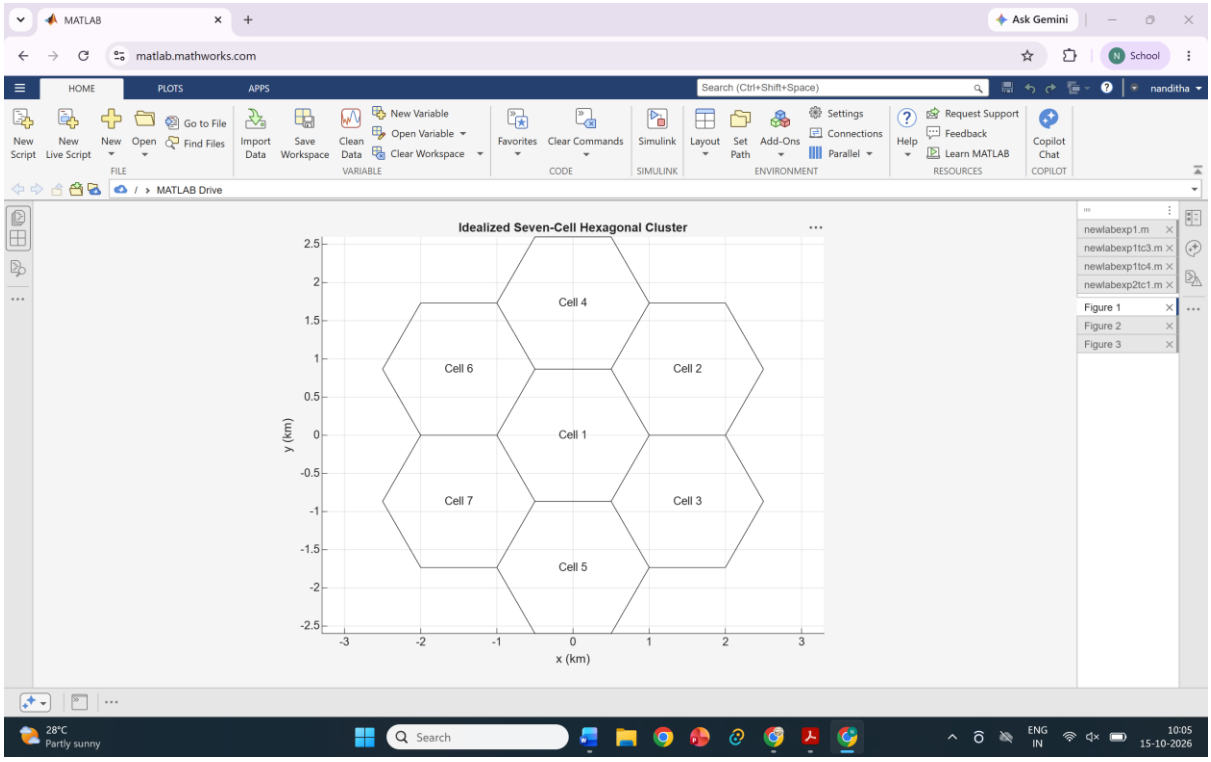
```
% Experiment 2: simplified hexagonal coverage and handoff
clear; clc; close all;
% Baseline assumptions
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
hexArea = @(R) (3*sqrt(3)/2) .* R.^2;
cellAreaKm2 = hexArea(radiiKm);
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
capacityTable = table(radiiKm(:), cellAreaKm2(:), cellCount(:), ...
repmat(usersPerCell,numel(radiiKm),1), estimatedUsers(:), ...
'VariableNames', {'Radius_km','CellArea_km2','EstimatedCells', ...
'UsersPerCell','IllustrativeUsers'});
disp(capacityTable);
% Plot a 7-cell cluster: central cell plus six neighbors
R = radiiKm(1);
centers = [0 0; 1.5*R sqrt(3)*R/2; 1.5*R -sqrt(3)*R/2; ...
0 sqrt(3)*R; 0 -sqrt(3)*R; -1.5*R sqrt(3)*R/2; ...
-1.5*R -sqrt(3)*R/2];
theta = (0:6)*pi/3; % flat-top hexagon vertices
figure; hold on; axis equal; grid on;
for k = 1:size(centers,1)
plot(centers(k,1)+R*cos(theta), centers(k,2)+R*sin(theta), 'k-');
text(centers(k,1),centers(k,2),sprintf('Cell %d',k), ...
```

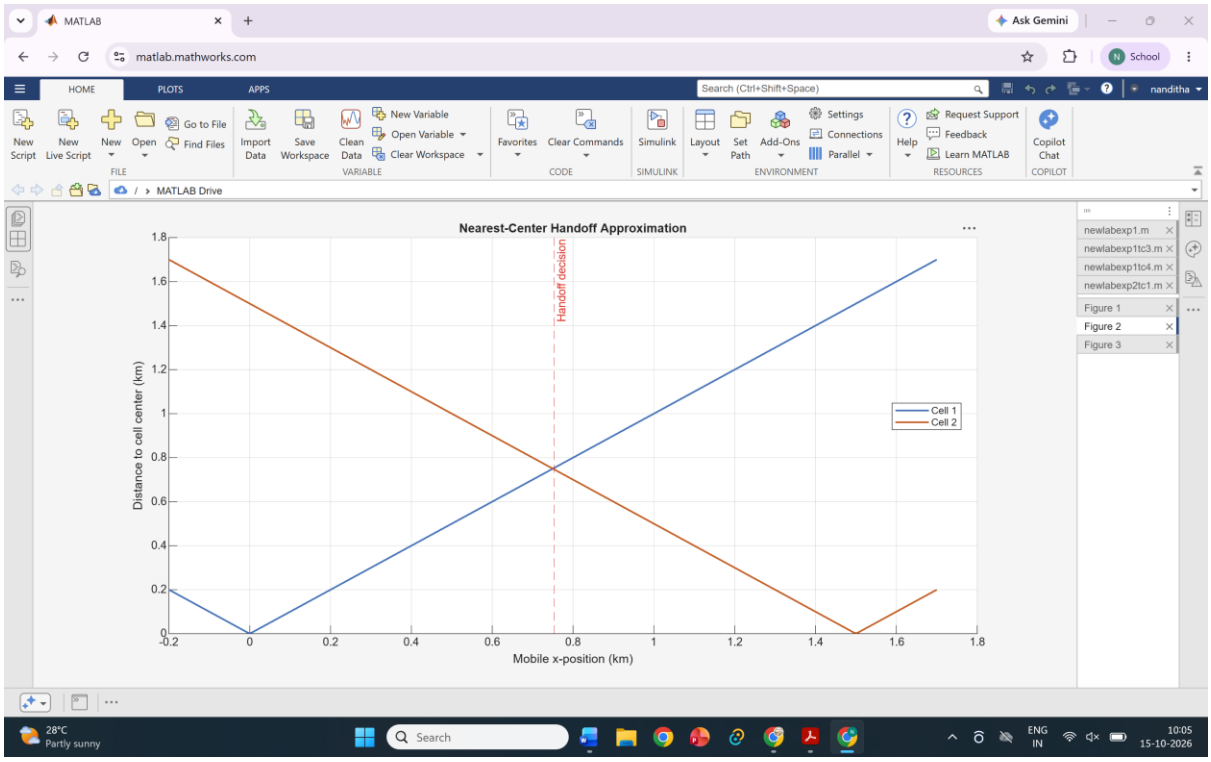
```
'HorizontalAlignment','center');
end
xlabel('x (km)'); ylabel('y (km)');
title('Idealized Seven-Cell Hexagonal Cluster');
hold off;
% Mobile moves across the boundary between centers at x=0 and x=1.5R
mobileX = linspace(-0.2*R,1.7*R,300);
mobileY = zeros(size(mobileX));
d1 = hypot(mobileX, mobileY);
d2 = hypot(mobileX-1.5*R, mobileY);
servingCell = ones(size(mobileX));
servingCell(d2 < d1) = 2;
handoffIndex = find(diff(servingCell) ~= 0,1,'first') + 1;
figure; hold on; grid on;
plot(mobileX,d1,'LineWidth',1.5);
plot(mobileX,d2,'LineWidth',1.5);
if ~isempty(handoffIndex)
xline(mobileX(handoffIndex),'--r','Handoff decision');
end
xlabel('Mobile x-position (km)'); ylabel('Distance to cell center (km)');
legend('Cell 1','Cell 2','Location','best');
title('Nearest-Center Handoff Approximation');
hold off;
figure; stairs(mobileX,servingCell,'LineWidth',1.5); grid on;
xlabel('Mobile x-position (km)'); ylabel('Serving-cell ID');
title('Serving Cell Along the Mobile Path');
if ~isempty(handoffIndex)
fprintf('Geometric handoff at x = %.3f km\n',mobileX(handoffIndex));
```

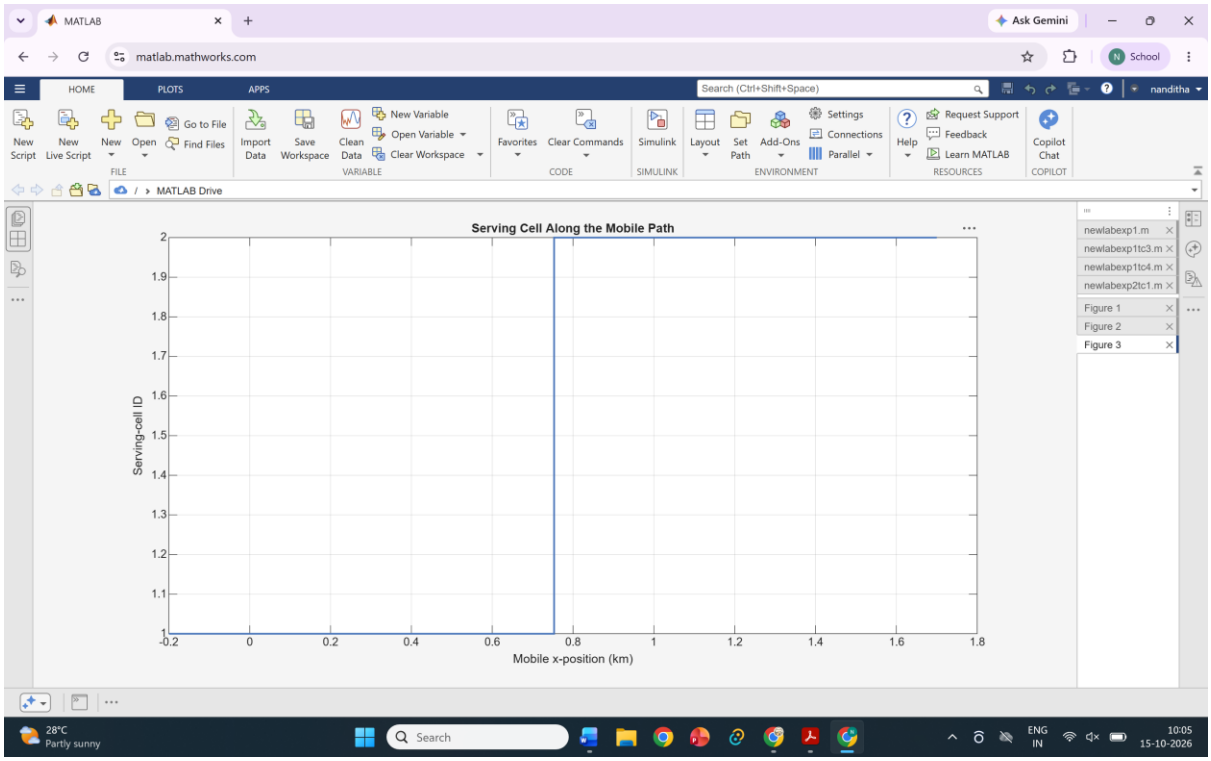
end

Result:


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HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
MATLAB Drive
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newlabexp1.m
newlabexp11tc3.m
newlabexp11tc4.m
newlabexp2tc1.m
Figure 1
Figure 2
Figure 3
Editor: 90%
UTF-8
Script
Ln 1 Col 1
28°C
Partly sunny
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APPS
MATLAB Drive
Idealized Seven-Cell Hexagonal Cluster
Cell 1
Cell 2
Cell 3
Cell 4
Cell 5
Cell 6
Cell 7
y (km)
x (km)
newlabexp1.m
newlabexp11tc3.m
newlabexp11tc4.m
newlabexp2tc1.m
Figure 1
Figure 2
Figure 3
28°C
Partly sunny
Search
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15-10-2026




## Results to Record

| Radius (km) | Cell area (km²) | Estimated cells | Users/cell | Illustrative users |
|---|---|---|---|---|
| 1 | 2.5981 | 2 | 30 | 60 |
| 0.5 | 0.64952 | 8 | 30 | 240 |

| Observation | Recorded result |
|---|---|
| Simulated handoff x-position (km) | 0.753 km |
| Effect of halving radius on estimated cells | |
| Effect of changed parameter | |

**Analysis Questions**

1. Why does halving the radius increase the idealized number of cells by approximately four

for a fixed area?

2. What assumption makes the estimated total user capacity scale directly with the number

of cells?

3. At what position does the nearest-center rule switch the serving cell, and why is that point

the geometric boundary?

4. Name two radio or network factors absent from this model that could shift or delay a real

handoff.

**Conclusion**

Summarize how cell radius changes the idealized cell count and assumed network capacity,

and report the simulated handoff location. State that the model uses ideal hexagons, an arearatio

count, fixed per-cell capacity and nearest-center handoff; it does not model RF

propagation, overlap, fading, load or handoff timers.